
Amazon CloudFront – Working, TTL, and Pricing

How Amazon CloudFront Works

CloudFront is a **content delivery network (CDN)** that serves your web content from **edge locations** close to your users to reduce latency and improve performance.

Workflow:

1. User Request:

A client (browser/app) makes a request to a CloudFront distribution (e.g., d123abc.cloudfront.net/image.jpg).

2. Edge Location Routing:

CloudFront routes the request to the **nearest Edge Location** (based on latency).

3. Cache Check:

- If the content is **cached** at the edge:
 -  Content is returned immediately (**cache hit**).
- If not cached (**cache miss**):
 -  CloudFront fetches the content from the **origin** (S3, EC2, ALB, or custom).

4. Response Caching:

- CloudFront caches the content at the edge with a **TTL (Time to Live)**.
- Subsequent requests during TTL duration are served from the cache.

5. Return to User:

Response is sent back to the user with much lower latency than fetching from origin.

Edge Locations and Regional Edge Caches

- **Edge Locations:**
Datacenters in global locations where content is served from cache.
 - **Regional Edge Caches (RECs):**
Intermediary caching layer between the origin and edge. Useful for large files or content requested infrequently.
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TTL (Time to Live) in CloudFront

TTL defines how long an object stays in the CloudFront edge cache before it expires and is re-fetched from the origin.

TTL Settings:

- **Minimum TTL:**
The least amount of time content stays in cache.
- **Maximum TTL:**
The longest duration the content can remain in cache.
- **Default TTL:**
The fallback time used if origin doesn't return **Cache-Control** or **Expires**.

You can control TTL using:

- **Cache behavior settings** in CloudFront.
- **HTTP headers** from the origin:
 - **Cache-Control:** max-age=3600
 - **Expires:** <timestamp>

⚠ Lower TTL → fresher content, but higher origin load

✅ Higher TTL → better performance, less origin traffic

💰 CloudFront Pricing Overview

CloudFront pricing is **pay-as-you-go** and **region-based**. Main pricing components:

Component	Description	Typical Cost
Data Transfer Out (DTO)	Data served to users from edge	Varies by region (e.g., \$0.085/GB first 10TB)
HTTP/HTTPS Requests	Charged per 10,000 requests	~\$0.0075 per 10,000 reqs
Invalidation Requests	First 1,000 paths/month are free	\$0.005 per path beyond
Field-Level Encryption	Extra charge for sensitive data	Based on usage
Lambda@Edge	Charges for compute + duration	~\$0.60 per 1M requests + GB-seconds

🔍 Cost Optimization Tips:

- Set longer TTLs for static content
 - Use compression (Gzip/Brotli)
 - Limit invalidation calls
 - Cache API responses where possible
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Quick Real-World Example

You host images in S3 and distribute them via CloudFront:

- Image size: 1 MB
- Users: 100,000 in a month
- Region: US
- All images cached at edge (no re-fetch)

Approximate Cost Breakdown:

- 100,000 HTTP requests → ~\$0.75
 - 100,000 MB ($\approx$ 100 GB) data transfer → ~\$8.50
 - **Total Monthly Cost** $\approx$ \$9.25 (basic usage)
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